

Assignment: Synthetic Survey Response Generator

Problem

Survey data collection is slow and expensive. Your task is to build a system that takes a survey as input and generates N realistic synthetic responses.

A good response is **plausible** (reflects how a real person might respond) and **coherent** (answers within a single response don't contradict each other).

What to Build

A program that accepts a survey definition and outputs N synthetic responses.

Your system must handle at minimum:

- Rating scales (1–5, NPS 0–10)
 - Single-choice questions
 - Open-text questions
-

Test Survey

Build and demo against this survey:

E-commerce Customer Satisfaction

1. Overall satisfaction? (1–5)
2. Likelihood to recommend? (0–10)
3. Category purchased? (Electronics / Clothing / Home / Other)
4. Was delivery on time? (Yes / No)
5. What could we improve? (open text)

Generate **200 responses**.

Deliverables

1. **Working code** — runs and produces output
 2. **A short write-up (max 1 page)** covering:
 - Your approach and why you chose it
 - How you'd measure whether your outputs are any good
 - One thing you'd do differently with more time
-

Evaluation

What we're looking for	Weight
Does it work?	30%
Are the responses realistic and coherent?	30%
Quality of reasoning in the write-up	25%
Code clarity	15%

Constraints

- Any language or model is fine
 - If you use an LLM, a single naive prompt is not enough — go further
 - Budget: stay within free tiers or under \$2 in API costs
-

Short, focused, leaves the approach entirely open. Want me to add a bonus stretch goal for students who finish early?